

# UAV Crop-Weed Segmentation for Precision Agriculture

## Goal

This project aims to train a deep learning model that segments UAV crop images into three pixel-level classes: crop, weed, and background/soil. The model supports precision agriculture decisions by identifying weed regions that could guide targeted monitoring or selective spraying.

## Data

I will use the public Kaggle **Crop-Weed Segmentation UAV** dataset:

<https://www.kaggle.com/datasets/bouhadjer/crop-weed-segmentation-uav-rgb-indices>

The dataset contains high-resolution UAV images of sorghum fields and segmentation masks with three classes: crop, weed, and background/soil. The original dataset contains approximately 6,300 annotated image patches generated from 19 UAV images. I will use the RGB images only, together with their corresponding segmentation masks.

I will use the provided test set for final evaluation, and split the training/validation data using a fixed random seed. The dataset is released for research and educational use under CC BY 4.0. Since the images are agricultural field images and do not contain people, the main ethics concerns are correct attribution, reproducibility, and avoiding overclaiming real-world deployment performance.

## Approach

This is a semantic segmentation problem. I will first build a simple encoder-decoder CNN baseline in TensorFlow/Keras. Then I will train a U-Net model, because U-Net is designed for dense pixel-level prediction and uses skip connections to preserve spatial detail.

The workflow will include loading image-mask pairs, resizing images, normalising RGB values, converting mask colours into class labels, training the models, and visualising predicted masks against ground truth masks.

## Success Criteria

The primary metric will be **Mean Intersection over Union (Mean IoU)** on the test set. I will also report Dice score and weed-class IoU, because detecting weed pixels is the most important practical goal.

The naive baseline is a majority-class prediction, such as predicting background/soil for all pixels. The learning baseline is a simple encoder-decoder CNN. The main success criterion is that the U-Net model improves over these baselines, especially on weed-class IoU.

## Risks & Scope

The biggest technical risk is class imbalance: background/soil pixels may dominate the masks, making weed pixels harder to learn. If needed, I will use class weighting, Dice-based loss, or focused error analysis to address this.

To keep the project manageable, I will not use vegetation indices, object detection, edge deployment, or advanced segmentation architectures unless the core RGB segmentation pipeline is complete. The final scope is RGB semantic segmentation with a baseline model, U-Net comparison, quantitative evaluation, and qualitative error analysis.